OP Engine

Numerical methods and their applicable models, outputs and conditions.

Method family	Models and methods	Outputs and conditions
Deterministic solvers	PDE / field, nonlocal, ODE Explicit, IMEX and implicit	Declared discretizations Observations, trajectories Accuracy and invariant checks
Stochastic solvers	Reaction networks Direct SSA, thinning Fixed / adaptive tau-leaping	Optional firing counts Independent NumPy SSA clocks
Schedule replay	Adaptive mesh discovery Frozen-schedule reevaluation	Selected deterministic paths Derivatives conditional on the recorded mesh

Tested NumPy, JAX and CPU Torch paths vary by method. Spatial and temporal accuracy need their own checks. GPU paths need separate hardware qualification.